CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (ME)

September 2018

ME-309 Design of Machine Elements-I

Date: 18.09.2018, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use of PSG design databook is permitted in examination, if required.

SECTION - I

Q - 1 Find out the numbers of the R10 basic series from 1 to 10. **[07]**

OR

State the general principles for the design of casting.

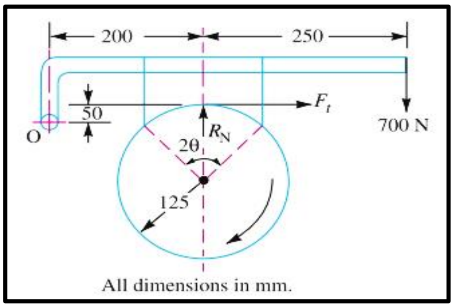
Q – 2 Determine the maximum, minimum and average pressure in a plate clutch when the axial force is 4 kN. The inside radius of the contact surface is 50 mm and the outside radius is 100 mm. Assume uniform wear. **[08]**

OR

Q – 2 1. State the characteristics of materials used for friction surfaces. **[06]**

2. State the different types of friction clutches. **[02]**

Q – 3 (a) 1. A single block brake is shown in Figure. The diameter of the drum is 250 mm and the angle of contact is 90°. If the operating force of 700 N is applied at the end of a lever and the coefficient of friction between the drum and the lining is 0.35, determine the torque that may be transmitted by the block brake. **[06]**



2. State the different types of brakes. **[02]**

(b) 1. Writedown advantages and disadvantages of Flat Belt drives over V Belt drives. **[06]**

2. What are the desirable properties of belt materials. **[06]**

OR

(b) Elaborate belt tensioning methods used to adjust belt tension. **[12]**

SECTION – II

Q-4 Differentiate between goodman, soderberg and gerber line with graphical representation. **[12]**

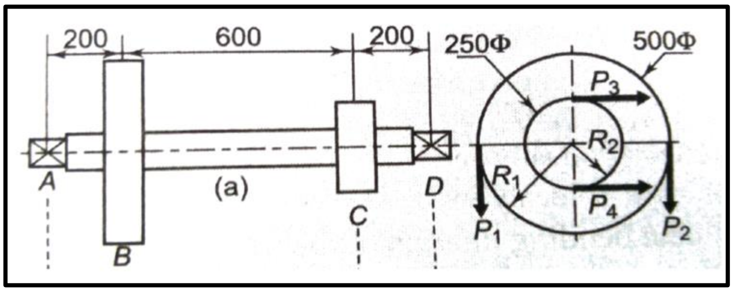
OR

Q - 4 (a) A forged steel bar, 50 mm in diameter, is subjected to a reversed bending stress of 250 N/mm^2. The bar is made of steel 40C8 (S_{ut} = 600 N/mm^2). Calculate the life of the bar for a reliability of 90%. Assume Ka = 0.44, Kb = 0.85, Kc = 0897. **[07]**

(b) Elaborate the factors affecting on endurance limit. **[05]**

Q - 5 (a) Define Power Screw and derive an equation for required torque to lift the load. **[05]**

(b) The layout of a transmission shaft carrying two pulley B and C and supported on bearing A and D is shown in fig. Power is supplied to the shaft by means of a vertical belt on pulley B, which is then transmitted to pulley C carrying a horizontal belt. The maximum tension in the belt on pulley B is 2.5 kN. The angle of wrap for both the pulley is 180^o and the coefficient of friction is 0.24. The shaft is made of plain carbon steel 30C8 (S_{yt} = 400N/mm^2) and the factor of safety is 3. Determine the shaft diameter on strength basis. **[10]**



OR

(b)
1. Derive an equation of strength for a sunk key with usual notations and sketch. **[05]**
2. Differentiate coupling and its types. **[05]**

Q - 6 Derive an equation for cylinder, which is made of brittle material. **[08]**

OR

Q - 6 An air receiver consisting of a cylinder closed by hemispherical ends is shown in Fig. It has a storage capacity of 0.25 m^3 and an operating internal pressure of 5 MPa. It is made of plain carbon steel 10C4 (S_{ut} = 340 N/mm2) and the factor of safety is 4. Neglecting the effect of welded joints, determine the dimensions of the receiver. **[08]**